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G 001 - GENERAL SYSTEM SPECIFICATIONS

1. GENERAL

The purpose of these system specifications is to define the system design and functional requirements, which fall within the Bidder's Scope of Work as defined in Part 4, Subsection 1.2.

It is the intent of this specification that fully functional systems shall be supplied by the Bidder suitable to achieve the specified targets for the project.

The total gross electrical power output of the Combined Cycle plant at site conditions shall be as specified in part 4, section 1. The service life as defined in ISO 3977, for the whole Combined Cycle plant shall not be less than 25 years.

The plant shall be able to operate under all prevailing ambient conditions at site, refer to Part 4 Subsection 1.1.

Design and guarantee conditions are shown in Part 5 Schedule 6.

The Maximum Continuous Rating of the Combined Cycle (Block-MCR) to be quoted and guaranteed shall be the rating achieved with the gas turbine generator operating at its base continuous rating.

The generating plant and auxiliaries shall be designed to ensure that trouble-free starts are achieved on a reliable basis throughout their design life. The generating plant shall also meet the following steady state and transient operating conditions:

- (1) Ability for the block to operate under auto control for sustained periods at minimum loads down to 50% of the given rated combined cycle for the given ambient temperature range.
- (2) To provide an initial steam turbine loading rate of not less than 3 % Block-MCR per minute. Following heat soak of Steam Turbine and for loads above 5 % Block-MCR, a loading rate of 5 % Block-MCR per minute for the complete Combined Cycle plant shall be available, without any restrictions.

- (3) All turbine generators are required to sustain full load rejection duties without tripping. The gas turbine generator is further required to remain operating at synchronous condition for sustained periods following load rejection and supporting its own auxiliaries without dependence on station supplies. The steam turbine generator (STG) may after an agreed time delay be tripped automatically under this no load condition when temperature stress conditions may arise.

The plant shall be designed so that this plant is entirely suitable for meeting the base load duty throughout its design life. The Bidder shall not make any subsequent claims should he consider additional or alternative designs to those specified are required to meet the specified duties.

2. TYPE OF PLANT

The Steam Turbine (ST) units shall be housed in fully weatherproofed ventilated turbine halls and shall be laid out generally in accordance with the Owner's General Arrangement drawings. The design of the HRSG, ST and ancillary equipment shall be fully proven by extensive, successful commercial operation at base load, and shall include the latest materials and experience with the type of machine and auxiliaries he is offering, giving installation dates running hours, type of duty, proven loading rates, etc. Where the Bidder's experience with particular components is limited, these items shall be listed in the Part 5 Schedule 3.2 "Deviation and Exceptions to Bid Documents".

The design of the installation shall limit the potential detrimental effects such as noise and pollution on the local environment to levels approved by the statutory regulations. Where limits are not defined by the Indonesian Regulations or this Technical Requirement, then the WHO Recommendations shall be the minimum standards applicable.

The plant should be designed with a high overall thermal efficiency. The unit will not always operate at its base load continuous rating and so consideration should be given to achieving high efficiencies at part load conditions without sacrificing operational flexibility.

In order to achieve maximum overall station efficiency and operational flexibility due consideration shall be given in the design of the combined cycle

block to enable the ST to have the capability to operate on variable pressure mode.

The heat cycle design can either include a dual or triple pressure design with reheat. The steam parameter for the different pressure-levels shall be proposed by the Bidder. A Steam Turbine exhaust moisture content of 12% shall preferably not be exceeded. The moisture content shall be stated in the bid together with the measures provided to guarantee absence of erosion during full and partial load down to 50%.

For the design of the HRSG both natural and forced circulation can be accepted.

The HRSG shall be vertical or horizontal flue gas flow.

3. OPERATING CONDITIONS

The combined cycle power plant will be operated primarily (at least for the first half of its design life) as a base load plant but shall be capable of alternate operating patterns. With the proposal heat balances shall be submitted as specified in Part 4, section 2.2.2.3. The plant will be operated on the block principle without connections to existing blocks.

The combined cycle plant Block 2 shall be controlled and supervised from the existing central control room (CCR) for Block 1 and Block 2. For Block 3 and Block 4 a new CCR shall be provided. Sufficient instrumentation shall be available in the CCR to identify any potential deterioration in the Add On plant performance and identify any fault, which would eventually cause a plant trip.

Normal start-up of the plant shall be fully automatic. The facility for fully manual start, loading and shutdown shall also be available, together with semi-automatic start: crank or pre-synchronizing or initial block loading.

The ST shall also be capable of being auto-synchronized and loaded.

Facilities for auto-synchronizing and step loading the combined cycle plant and manual synchronizing shall be provided.

Provision for further loading after above target load to be controlled from Load Dispatching Center (LDC) shall be provided as specified in Part 4 Section 3, I-100.

3.1 Start-Up Requirements and Load Change

The, HRSG and ST in combined cycle plant shall be capable of being started, warmed, run to speed and fully loaded in the shortest possible time. The time to achieve full design conditions for each item of plant from a cold, warm and hot initial condition shall be stated in the proposal in the form of schedules and curves.

The ST shall be provided with a by-pass system to improve operational flexibility under start up, load changing and shut down conditions, and to enable the HRSG to be brought on stream independently.

During a combined cycle plant start up the auxiliary plant (including the condenser vacuum system and the gland steam shaft sealing system for the turbine rotor) shall be capable of satisfactory and automatic operation to achieve a rapid start up. The operation of the ST by-pass system during the start up sequences shall also be automatic and designed to prevent over-rapid-heating or pressurization of any plant item.

The combined cycle plant shall be designed to be capable of responding to the load demands imposed by the system and dictated by the governor control system. The Bidder shall advise response times to these cyclic load demands for combined cycle modes.

3.2 Variable and Constant Steam Pressure Operations

The ST and HRSG shall be designed to be capable of both constant pressure and variable pressure operation from 100% down to about 50% load.

Since a considerable proportion of the life of block operation will be at outputs below the design capability, high part load efficiency is regarded as important and essential. In addition, frequent starts and stops may be called for associated with minimum start-up and shut-down periods. Minimum fuel use under these start/stop conditions is also regarded as important.

The ST and HRSG shall be capable of operating continuously on fully automatic control under variable pressure mode of operation at all loads. Automatic control arrangements and anticipated operating regimes are further detailed in the Instrumentation and Control section of this Specification.

The 100% bypass system set points and sliding pressure of ST above 50% load shall be closely coordinated to support the frequency variations however the primary response will be based on GT and ST will follow the GT in case of frequency variations and will set revised ST sliding pressures and bypass set points.

The Bidder shall comment on any special requirements, precautions or limitations associated with these modes of operation. Particular attention shall be paid to the design and operational requirements of turbine control gear, pressure unloading gear and protective devices under the variable pressure mode of control.

3.3 Load Rejection

In case of load rejection by the Java/Bali grid system under certain emergency circumstances, the plant can be sustained at reduced load to maintain unit auxiliaries (House Load Operation by Gas Turbine Generators) without black start capability.

Resynchronization to the grid and recovery to full load shall be required as soon as possible after the cause of the load rejection has been cleared. ST may be tripped under the house load operation, but the HRSG shall be kept in operation if gas turbine is still running. Under these circumstances, the steam turbine bypass system should operate accordingly to regulate the steam pressure and prevent lifting of the safety valves. The gas turbine exhaust damper shall not be used for this purpose.

The GT generator shall be capable of remaining in operation for an indefinite period at a load to maintain unit auxiliaries. When the load demand is returned the plant shall be capable of returning rapidly to the original combined cycle status. Any limitations on rapid re-loading after such a load excursion shall be stated in the proposal.

3.4 Operation on Failure of A.C Supplies

Following a severe Java/Bali grid system disturbance resulting in tripping of the remained GT generator leading to the loss of all AC power supply to the unit auxiliaries, the plant shall be capable to be placed to safely shutdown of the unit by placing onto barring gear operation. During this period essential services for GT and ST units shall be fed from DC supply and emergency diesel generators to enable the plant to be rapidly re-loaded.

4. PLANT LAYOUT AND DESIGN

Subject to the Owner's approval and to ensure a satisfactory overall plant layout, the Bidder shall be responsible for the satisfactory arrangement and layout of all Add On plant covered by this Specification.

The type, arrangement and layout of all plant and buildings shall be generally in accordance with the proposed plant layout as indicated on the Specification Drawings and shall also conform to the Technical Requirements. The Bidder shall state the dimensions of the ST Hall and any other buildings such as the central control room of Block 3 and 4, and ST electrical annex areas which may in whole or part accommodate equipment supplied under this contract. Cabling requirements shall also be identified by the Bidder for those areas.

The size of access routes and lay-down areas shall not be less than that shown on the Bid document drawings. The lay down areas required for a major overhaul of the plant shall be shown in a drawing submitted with the bid.

The design and arrangement of plant, particularly in the ST building shall be such that the extent of any hazardous areas is minimized

The ST buildings shall be adequately ventilated and ST shall be located such that, with the possible exception of electrical equipment mounted on the ST itself, electrical equipment within the ST buildings shall be unclassified.

The Bidder shall be responsible for ensuring that the design of the ST Halls, electrical annexes, any cable or duct routes or any civil work within this Contract does not result in water entering or accumulating in any areas where mechanical, electrical or instrumentation and control equipment are installed.

All cable trenches shall be designed with a fall to a sump with arrangements for emptying of the sump.

The proposal shall include a comprehensive and fully detailed section on maintenance and servicing, suitably illustrated with diagrams and drawings and justification for the maintenance or service life of the plant items described based on previous experience with plant of the same type.

In particular, the recommended procedure for carrying out the following operations of maintenance works shall be described:

- (1) Repair of HRSG casing, headers and tube elements.
- (2) Removal, repair and replacement of gas path dampers, seals and mechanism.
- (3) Removal and replacement of the ST rotor, blading and bearings.

The Bidder shall provide sufficient space for maintenance of plant equipment and shall guarantee that all maintenance works specified above could be performed easily.

5. SATISFACTORY ACCESS TO PLANT

In developing his plant layout, the Bidder shall ensure that satisfactory access for maintenance, inspection and operation is achieved to all Add On equipment and valves. Suitable access galleries, ladders, stairways and floors with appropriate hand-railing and the like shall be provide for this purpose. Only in exceptional cases will mobile or portable platforms be acceptable.

6. APPEARANCE COVERS FOR STEAM TURBINE GENERATORS

Turbine-generator appearance covers shall be provided in accordance with the Bidder's standard design and also where required for noise attenuation purposes.

Where fitted, the appearance covers shall be of pleasing appearance, suitably stiffened and secured to avoid thermal distortion. They shall be resistant to corrosion and mechanical damage.

For the purpose of reducing maintenance outage periods to a practicable minimum, the appearance covers shall be designed to be erected in sections, each section being readily removable. The sections shall be strengthened to facilitate removal and replacement without damage and arranged so that when

sections are removed, instruments, instruments panels, gauge boards and connections need not be disturbed.

Adequate door access shall be provided to the inside of the covers. Where necessary, suitable lighting shall be provided under the covers arranged to produce an adequate lighting level that is compatible with safe working and safe access.

All local instrumentation gauges, oil sight glasses, slip-rings and the like, local to the turbine generator are to be visible from the outside of the appearance covers.

7. MECHANICAL

The Bidder shall design all the systems based on the following:

7.1 Mechanical System General Requirements

The design conditions of all systems shall be based on the most severe conditions from the heat balance diagrams and/or other design conditions. In addition to the equipment specifications mentioned specifically in the individual system functional requirements, the following specifications, or portions thereof are also applicable to the systems described herein:

- M-001 : General Specification for Piping
- M-002 : Pipe Supports and Hangers
- M-003 : Thermal Insulation and Lagging
- M-004 : Valves
- M-005 : Control Valves
- M-006 : Horizontal Centrifugal Pumps
- M-007 : Vertical Centrifugal Pumps
- M-008 : Vertical Sump Pumps
- M-009 : Metering Pumps
- M-010 : Miscellaneous Heat Exchanger
- M-011 : Miscellaneous Hoists and Trolleys
- M-012 : Shop Fabricated Tanks

M-014 : Inspection and Tests

7.2 Mechanical System Functional Requirements

7.2.1 Exhaust Gas System

Reference Specification

M-305 : Ducting, Damper, Main Stack and By Pass Stack

Function Requirements

- (1) To provide a flow path for combustion turbine exhaust gases from the gas turbine exhaust to the heat recovery steam generator inlet duct work or by pass to the atmosphere
- (2) To control the noise level of GT exhaust gas flow to the acceptable noise requirement
- (3) To control pressure upstream of the heat recovery steam generator damper.
- (4) To isolate HRSG from the gas turbine and allow gas exhaust through by pass stack
- (5) To allow personnel to enter HRSG, when GT is operating.

7.2.2 Boiler Vents and Drains

Reference Specification

M-301 : HRSG and Auxiliary Equipment

Function Requirements

- (1) To allow venting of nitrogen and air from the HRSG when it is being filled with water.
- (2) To vent steam during start-up
- (3) To allow continuous steam drum blow-down of the different pressure systems to maintain acceptable boiler water chemistry.
- (4) To provide a mean of disposing of the steam generator water when it is being drained.

- (5) To provide a way for relieving the pressure within the steam generator steam and water circuits in the event that an excessive pressure condition should occur.
- (6) To facilitate chemical cleaning, without requiring separate chemical cleaning connections.

7.2.3 Main Steam

Reference Specifications

- M-301 : HRSG and Auxiliary Equipment
- M-001 : General Specification for Piping

Function Requirements

To route high-pressure, the intermediate pressure (as far as applicable) and low-pressure steam from the HRSG steam outlets to the steam turbine.

7.2.4 Turbine Bypass

Reference Specifications

- M-301 : HRSG and Auxiliary Equipment
- M-001 : General Specification for Piping
- M-302 : Turbine Bypass System

Function Requirements

- (1) To reduce time for steam generator/steam turbine unit start-up (during cold, warm and hot start-up) and improve matching of steam and metal temperature, thus reducing steam turbine rotor and cylinder thermal stresses.
- (2) To allow the steam generator to continue operation at reduced load after a system or a turbine load rejection. In the case of a system load rejection, gas turbines shall maintain house load. In case of a full load rejection safety valves shall not blow off. The turbine bypass shall be sized by the Bidder to meet the above criteria.
- (3) To allow the steam generator to operate independently from the Steam Turbine generator up to 100% of HRSG load.

7.2.5 Blow Out/Chemical Cleaning

Reference Specification

M-301 : HRSG and Auxiliary Equipment

Function Requirements

To provide facilities for removing oil, grease, paint, rust inhibitors, weld spatter, rust, mill scale and other contaminations which may be present in the HRSG and the main steam piping before initial operation.

- (1) HRSG
- (2) Auxiliary steam lines
- (3) The main steam lines
- (4) The turbine bypass steam system

7.2.6 Auxiliary Steam System

Reference Specification

M-304 : Auxiliary Steam System

The Block 2 auxiliary steam system shall be connected to the Block 1 auxiliary steam system.

The Block 3 & 4 auxiliary steam system shall be connected to the new desalination plant.

The new Auxiliary Boiler shall supply Auxiliary Steam to the new desalination plant (2 x100% Desalination Plant).

The existing Auxiliary Boiler shall supply Auxiliary Steam to the new desalination plant (2x100% Desalination Plant).

7.2.7 Turbine Vents and Drains System

Function Requirements

The turbine vents and drains shall provide venting and draining facilities for the various items of equipment associated with the main steam turbine. The purpose of the system is:

- (1) To remove air and non-condensable gases from steam related equipment and vent them to atmosphere.
- (2) To remove air from oil tanks in order to prevent condensation.
- (3) To provide means of draining all items of equipment.
- (4) To route water drains to the floor drains or to condenser, as applicable
- (5) To route oil/water mixture to the oil separator.

7.2.8 Turbine Lubricating Oil System

Reference Specifications

M-201 : Steam Turbine

M-209 : Station Oil Purifier

Function Requirements

- (1) To provide cleaned lubricating oil in sufficient quantity and at proper pressure for proper lubrication of turbine bearing during start-up, idling operation at all loads and shut-down of the, steam turbine and the generators.
- (2) The system can also, supply oil for relay oil, governing oil, jacking oil for all turbine and associated systems. A separate control oil system is acceptable
- (3) The system shall have oil cleaning and purifying capabilities without disturbing turbine operations.

7.2.9 Turbine Gland Steam Sealing System

Reference Specification

M-201 : Steam Turbine

Function Requirements.

- (1) The turbine steam sealing system shall properly seal the clearances between the shaft glands and the shaft. The system shall protect against leakages during start-up, normal operation and shutdown.
- (2) All motor operated valves shall be controlled from the central control room.

7.2.10 Condensate System

Reference Specifications

- M-204 : Condensate Pumps
- M-205 : De-aerator
- M-202 : Condenser
- M-405 : Chemical Injection System

Function Requirements

- (1) The purposes of the condensate system is:
 - a. To condense the exhaust steam from the turbine.
 - b. To pump the condensed steam (condensate) from the main condenser.
 - c. To increase the condensate temperature in the de-aerating heater.
 - d. To inject chemicals to protect piping, and the HRSG from corrosion and scale build-up.
 - e. To remove residual and other gases by de-aeration.
 - f. To provide condensate to the suction side of the boiler feed pumps.
- (2) In addition, the condensate system shall supply condensate to the following equipment:
 - a. Turbine exhaust hood spray headers
 - b. Desuperheater for turbine gland steam and LP auxiliary steam (after PRV)
 - c. Miscellaneous drain tank overflow seals
 - d. Vacuum breaker seals
 - e. Chemical solution tanks, when required
 - f. Condensate pumps seals
 - g. Sealing water system for the fittings in the systems under vacuum
 - h. Closed cooling water head tank (ST building)
 - i. Bypass systems attemperating water

- (3) To ensure that, at low unit loads, the flow through the gland steam condenser and the condensate pumps is above the minimum required, condensate shall be recirculated from downstream of the gland steam condenser back to the main condenser via properly selected control valve. The amount of condensate recirculated shall be determined by the greater of the minimum requirements of the gland steam condenser and the condensate pumps.
- (4) Condensate flow to the deaerator shall be regulated by two pneumatic control valves operating based on set points derived from hotwell level control loop. The deaerator level control shall be done with the make up valves to condenser.
- (5) During steady state conditions, the condenser hot-well and deaerator level shall be maintained substantially constant by the condensate flow control system.
- (6) Both deaerator and condenser hotwell level control loops shall operate in conjunction with each other. The condensate discharge control valve, normal and emergency make up water valves and excess condensate dump valve shall function based on the different level switches in condenser hotwell and deaerator. Both the condensate hotwell level control and deaerator level control shall derive the set points of the condensate discharge control valve, open close binary signals of normal and emergency make up water valves and condensate dump valve to keep the deaerator and condensate level in control in case of variations. The condenser hotwell level control shall be faster compared to deaerator considering the storage volumes difference between hotwell and deaerator. The condensate dump valve shall only operate at very high levels and can be connected to waste water system
- (7) Before the turbine shaft can be sealed, the condensate system will be put into operation in order to provide cooling water to the gland steam condenser. During cold starts of the turbine, condensate shall be provided to desuperheat the auxiliary steam (if available) used for gland sealing.
- (8) Configurations of three (3) x 50% condensate extraction pumps shall be provided. Two (2) condensate pumps shall be used for normal operation and the third pump shall be provided for automatic standby

When the working condensate pump is tripped, the standby pump shall automatically start. Seal water to the pumps shall be supplied from the pump discharge header alternatively a self-sealing plan shall be used for CEPs as per API code.

7.2.11 Feedwater System

Reference Specification

M-303 : Boiler Feed Pumps

Function Requirements

The function of the feedwater system is to supply feedwater from the deaerator to the different pressure systems of the HRSG.

(1) Boiler Feed Pumps

Each boiler feed pump shall be furnished with a recirculation line and an automatic recirculation cum check valve to ensure that the pump minimum flow requirement is satisfied. The recirculation lines shall individually be routed from the pump discharges to the de-aerating heater storage tank.

The recirculation cum check valve shall open automatically when the pump discharge flow is less than the minimum flow.

Each of the 3 x 50% feedwater pumps shall be directly driven by an electrical motor. In normal operation two (2) pumps shall be in service, and one (1) pump on standby. In the event of a pump failure, this standby pump shall be run in automatically. Either speed controlled motors or constant speed motors can be applied.

(2) Feedwater Flow Control

During start-up and low load the feedwater flow control system shall be single element using drum level and will modulate the control valve in the feedwater line.

When an increase in load is required, the feedwater control system shall automatically switch to three element control system operating on the function of drum level, steam flow and feedwater flow.

7.2.12 Turbine Room Vents and Drains

Reference Specification

M-201 : Steam Turbine

Function Requirements

- (1) The purpose of the turbine room vents and drains system is:
 - a. To remove air and non-condensable gases from equipment in the turbine room by using vacuum equipment or venting continuously to condenser or atmosphere.
 - b. To route discharge from the safety valves to atmosphere or floor drains.
 - c. To route water from turbine room equipment drains to floor drains.
 - d. To route oil/water mixture from turbine room equipment to oil separator.
 - e. To route miscellaneous drain tank discharge to condenser.
- (2) Condenser Vacuum System
 - a. Condenser vacuum will be established by using steam jet ejectors or water ring pumps to evacuate air from the steam side. Both hogging ejectors and normally continuously operating ejectors shall be started for hogging until the condenser vacuum reaches the value specified in Part 4, Specification M-202. The hogging ejectors shall then automatically switch off. The ejectors shall be operated from the central control room. Function of water ring pumps as far as applied shall be similar.
 - b. If the operating ejectors cannot maintain the condenser vacuum required, the hogging ejectors shall automatically switched on to utilize its full air removal capacity.
- (3) Miscellaneous Drains Tank
 - a. All external drains in the steam water loop in the turbine room operating during start up shall be flashed in this atmospheric drain vessel. The steam shall be discharged to atmosphere and the condensate shall be collected in the vessel.

- b. A condensate transfer pump shall be provided to discharge the condensate from the vessel. In case of high alarm level the transfer pump shall be switched on from the central control room to drain the condensate to the waste water system. Low level in the tank shall trip the pump.

(4) Turbine Room Stand Pipe

- a. All turbine internal drains shall be drained to the flash tank connected to the condenser. The drains system shall meet the requirements of Turbine Drains Protection (TDP) code.

7.2.13 Circulating Water System

Reference Specifications

- M-206 : Circulating Water Pumps
- M-207 : Circulating Water Screens
- M-208 : Circulating Water Piping

Function Requirements

- (1) The function of the CW system is to supply cooling water to condenser and the closed cooling water heat exchanger of the steam plant.
- (2) CW pumps and butterfly valves at CW pumps discharge will be started from the Central Control Room. Requirements of starting sequence are indicated in Specification M-206, CW Pumps.
- (3) 2 x 50% CW pumps per combined cycle block shall be sized for steam plant cooling duty.
- (4) For normal manual shutdown, circulating water pumps shall be equipped with delay devices to prevent shut down of the pump motor until the CW discharge valves are closed to an appropriate extend to prevent water hammer problems.

7.2.14 Chemical Injection System

Reference Specification

- M-405 : Chemical Injection System

Function Requirements

To minimize scaling and corrosion within the steam generator and steam generator tubes by feeding chemicals to the condensate and feedwater cycles.

7.2.15 Pressurized Air System

Reference Specifications

M-410 : Service Air System

M-411 : Instrument Air System

Function Requirements

(1) Normal Operation

Each air compressor control cabinet shall be provided with a three position selector switch with OFF/STANDBY/RUN positions. When RUN is selected, the compressor shall run continuously on load/unload cycle to maintain the pressure in the system. When STANDBY is selected, the compressors shall not start unless the header pressure falls to a preset value. If started, the compressor shall continue to run on load/unload cycle.

The STANDBY start pressure sensing device shall be located on the air receiver discharge header.

(2) Abnormal Operation

Back pressure control valves shall be utilized in the service air system which, in the event of insufficient air being available to meet the instrument air requirements shall close, thereby making more air available to the critical instrument air system.

These valves shall be located on the supply line to the service air ring distribution system.

(3) Pressurized air system complete with all the auxiliaries shall be designed .to supply instrument and service air to the whole plant including auxiliary plant outside the main building.

(4) Instrument air system shall have a pressure dew point of - 40°C.

7.2.16 Fire Detection and Protection System

Reference Specification

M-420 : Fire Protection and Detection System

Function Requirements

- (1) The Station Fire Fighting and Protection (FP) System is intended to minimize damage caused by fires. Early warning fire detection systems shall be provided such that by pursuing corrective action major fires can be avoided.
- (2) The fire detection shall be provided for all major equipment and relevant plant areas.
- (3) The system shall be new firefighting system consisting of one diesel driven fire pump, one (1) motor driven pump and one (1) jockey pump located in Fire Fighting Pump House.
- (4) The system shall be kept under pressure by the existing jockey pump. If the existing motor driven pump trips or pressure is lost in the FP system, existing diesel-driven pump shall start automatically.

7.2.17 Nitrogen System

Function Requirements

- (1) The nitrogen system shall supply nitrogen gas to fill the identified equipment during extended periods of standby or outage, thereby preventing corrosion due to the ingress of air.
- (2) The bottle storage area and the gas pressure reducing station shall be sized to provide sufficient nitrogen for the whole Combined Cycle block.
- (3) Flexible couplings shall be supplied at the equipment supply connection.

7.2.18 Closed Cooling Water System

One (1) independent closed cooling water system shall be provided for each Steam Turbine for block 2, 3 & 4.

Function Requirements:

- (1) General

The closed cooling water system (CCW) shall use demineralized water in a closed loop to cool all the equipment requiring cooling water.

An atmospheric, overhead tank shall maintain suction head to the CCW pumps and accommodate thermal expansion and contraction of system fluid.

(2) Steam Turbine CCW

For normal operation one (1) full size CCW pump shall be used to circulate cooling water through one (1) full size CCW heat exchanger and through the equipment coolers. A second full sized pump and heat exchanger shall serve as standby.

Normal make-up to the CCW head tank shall be taken from the make up water storage tank. The amount of make-up water shall be regulated to maintain a constant water level in the head tank. To fill the tank initially, the make up pumps shall be used.

A chemical dosing tank will be used periodically to feed the required amount of chemical into the CCW system by the manual opening of the valves in the tank line and allowing a recirculation flow of cooling water from the CCW pump discharge to the suction.

(3) Demineralized Water System

Reference Specification

M-401 : Demineralized Water System

Function Requirements

A suitable de-mineralized water system shall produce specified water quantity and quality required for the satisfactory operating of Combined Cycle Plant.

7.2.19 Waste Water Treatment System

Reference Specification

M-404 : Waste Water Treatment System

Function Requirements

Waste water treatment system shall treat the waste water discharged from Muara Tawar CCPP wastes to a quality required to meet the specified guide lines.

7.2.20 Sampling Rack System

Reference Specification

M-406 : Sampling Rack System

Function Requirements

A sampling system shall be provided to take samples for continuous measurements and laboratory analysis from the water/steam cycle and auxiliary systems. The analysis results obtained shall be used to assess the chemical condition of the operating fluids. The samples shall be cooled and condensed as necessary in the sampling equipment. Precision control valves shall serve to reduce the system pressure and to set a constant sample flow rate in order to obtain representative samples.

7.2.21 Ventilation and Air Conditioning System

Reference Specification

M-421 : Ventilation and Air Conditioning System

Function Requirements

- (1) Control building shall be provided with suitable air conditioning for proper work of control equipment and accessories.
- (2) Miscellaneous buildings shall be provided with air conditioning and ventilation for proper operation of equipment, and/or comfortable condition for operation and maintenance personnel including the administration staff.

7.2.22 Domestic Water Supply System

Reference Specification

M-403 : Raw Water Tank System

Function Requirements

To supply chlorinated service water to an elevated storage tank for domestic cleaning purposes like showering, handwashing, toilet flushing etc.

7.2.23 Service Water System

Reference Specification

M-403 : Raw Water Tank System

Function Requirements

The service water system shall provide service water to various systems within the power plant.

7.2.24 Desalination Plant

Reference Specification

M-440 : Evaporative Desalination Plant

Function Requirements

To desalinate seawater by means of a multistage flash evaporative type (MSF).

7.2.25 Hydrogen Plant

Reference Specification

M-441 : Hydrogent Plant

Function Requirements

Cooling for Steam generator Block 3 and 4

8. ELECTRICAL AND INSTRUMENTATION & CONTROL

8.1 Purpose

It is the intent of this Specification that safe, reliable and efficient operation of plant shall be achieved by the Bidder. The Bidder shall furnish all necessary materials, components and assemblies required to functionally support the Bidder's work. The purpose of these following sections is to define the

electrical and I&C system functional requirements which fall within the scope of work as defined in **Part 4 Subsection 1.2 "Summary of Works"**.

8.2 Electrical and I&C System General Requirements

The design conditions of all equipment shall be based on the most severe operating conditions as specified in **Part 4, Subsection 2.16 "Environmental Consideration"**, including the effects of exposure to intense sunlight on the temperature within outdoor race ways, enclosures and conduits, the degradation of cable insulation in outdoor cable trays, and the increase in ambient temperatures of operating equipment, etc. The electrical and I&C design shall be based on the applicable standards and in accordance with proven design principles and practice and to these specifications. Electrical and I&C equipment made to national or industry standards equal to or better than those specified in the equipment specifications shall also be acceptable. The Bidder shall prove the equality of substitute standards to the satisfaction of the Owner.

Power generated from the steam turbine unit generator shall be delivered to the 500 kV grid through the generator circuit breaker, the generator transformers and 500 kV circuit breakers . Power supply to the units auxiliaries shall be taken from the generator during its operation through the unit auxiliary transformers. Power supply for normal startup and shutdown for the plant shall be taken from 500 kV Grid through generator transformers, and unit auxiliary transformers with the generator circuit breakers open.

One (1) Emergency Diesel Engine Generator shall be installed for block 2 , **block 3 & 4** to provide alternative power to 400 V essential buses for essential electrical loads and services if there is no power from the normal buses.

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In general, two (2) power sources of bus design for the auxiliary power distribution shall be followed, with the components of each bus kept physically and electrically segregated.

The following specification or portions thereof are applicable to the systems described herein

(1) General

E-001 : General Electrical Specifications

E-002 : Test and Inspection

(2) Generator and Main Output System

E-101 : Generators

E-102 : Isolated Phase Bus Ducts

E-103 : Generator Circuit Breakers

(3) Plant Electrical

E-201 : Medium Voltage Switchgears

E-202 : Non Segregated Phase Bus Ducts

E-203 : Power Distribution Center (PDC) Transformers

E-204 : Power Distribution Center (PDC) Switchgears

E-205 : Motor Control Centers

E-210 : Low Voltage Motors

E-211 : Medium Voltage Motors

E-220 : Station Batteries and Chargers

E-221 : DC Distribution Panels

E-222 : Uninterruptible Power Supply

E-223 : Emergency Diesel Engine Generator

E-230 : Cable and Wires

E-231 : Cable Tray System

E-232 : Conduits and Fittings

E-233 : Underground Electrical Ducts

E-240 : Outdoor and Substructure Grounding

E-241 : Building and Equipment Grounding and Lightning Protection

E-250 : Unit Metering and Protection

E-260 : Circulating Water System Cathodic Protection

- E-261 : Cathodic Protection for Field Piping and Buried Structures
- E-270 : Lighting
- E-271 : Communication System
- E-272 : CCTV system
- E-280 : Panel Design
- (4) Power Transformers
 - E-301 : Oil Immersed Power Transformers
 - E-302 : Generator Transformers
 - E-303 : Unit Auxiliary Transformers
- (5) Switchyard
 - E-400 : Substation and Appurtenances.
 - E-401 : Air Insulated Bus Work.
 - E-410 : High Voltage Circuit Breakers
 - E-411 : Disconnecting and Grounding Switches
 - E-420 : Voltage Transformers
 - E-421 : Current Transformers
 - E-430 : Surge Arresters
 - E-440 : Substation Protection Relaying System
 - E-450 : AC/DC Distribution System
 - E-460 : Substation Control System
 - E-461 : Substation Monitoring System
 - E-471 : Dispatch System Interfaces
 - E-481 : 500 kV Compressed Gas Insulated Bus
- (6) Instrumentation and Control
 - I-001 : Instrumentation and Control
 - I-002 : Local Control Panels
 - I-003 : Local Mounted Instrument Panels
 - I-005 : Distributed Control System

I-100 : Plant Control System

I-200 : Plant Monitoring System

8.3 Electrical and I&C System Functional Requirements

8.3.1 Unit Metering and Protection

Reference Specifications

E-001 : General Electrical Specification

E-201 : Medium Voltage Switchgears

E-204 : Power Distribution Center Switchgears

E-205 : Motor Control Centers

E-250 : Unit Metering and Protection

Function and Design Requirements

- (1) The two (2) basic functions of the relaying system are to detect and protect against dangerous or abnormal operation or conditions in power plant equipment. The design of the auxiliary power system includes the responsibility for the furnishing and coordination of all protective relaying including the 400 Volt Motor Control Centers.
- (2) The metering equipment provides information required by the operator and records information needed for determining revenue and for maintenance purposes.
- (3) The minimum relaying and metering requirements to implement the above functions are shown in detail on the referenced drawings. General functional descriptions and specifications for the major components comprising the protective relaying are included in the referenced specifications.

8.3.2 Medium Voltage Auxiliary Power Distribution System

Reference Drawings

Key Single Line Diagrams

Reference Specifications

E - 001 : General Electrical Specification

- E - 201 : Medium Voltage Switchgears
- E - 202 : Non Segregated Phase Bus Ducts
- E - 203 : Power Distribution Center Transformers
- E - 250 : Unit Metering and Protection Panels

Function and Design Requirements

The function of this system is to provide a means of distributing auxiliary power at 6.3 kV ST Switchgear with the necessary coordinated relaying required to protect the loads it services.

- (1) The auxiliary power system shall supply AC power to large motors (>150 kW) through 6.3 kV switchgear and to low voltage auxiliaries through the step-down transformers in the 400 V Power Distribution Centers (PDCs).
- (2) For normal operation on ST Switchgears, the medium voltage (MV) 6.3 kV power will be obtained by stepping down the generator voltage through the gas and steam turbine unit auxiliary transformers. While during startup and shutdown, the auxiliary power will be obtained by stepping down the 500 kV off site power through generator transformers and unit auxiliary transformers (UAT) with two-winding type by opening Generator Circuit Breakers.
- (3) The low voltage side of the UATs will be connected to MV switchgears using non segregated phase bus ducts. The UATs will be located outdoor and will be oil cooled ONAN or ONAN/ONAF rated. It shall be provided with fire protection and also with fire walls as required.
- (4) The transformers and MV switchgear will have instrumentation and protection equipment as required.
- (5) The MV auxiliary power system will be designed to meet the power requirements of all the plant auxiliary loads during normal operation as well as during start-up, shut down, UAT outage, etc. The normal operating voltage for MV switchgear will be 6.3 kV (nominal 6 kV).
- (6) Each 6.3 kV Switchgear assembly will supply a separate and independent load group during normal operation. The loads on the 6 kV

bus will be so assigned that if normal power from the relevant unit auxiliary transformers is interrupted, the plant auxiliary equipment will be supplied from other unit auxiliary transformer.

- (7) In general, the majority of the auxiliary loads which are considered essential for immediate operation of the unit will be fed from the unit auxiliary system. Vital loads which are essential for the safe shutdown at the unit for control and instrumentation will be fed from the 400 V Essential PDCs, DC Distribution Panels and from the Uninterruptible Power Supply (UPS).
- (8) The 6.3 kV Switchgear will normally supply large motors (>150 kW) and PDCs. The smaller auxiliary loads will be supplied from the 400 V PDCs and the 400 V motor control centers.
- (9) **An "automatic fast transfer" of the 6.3 kV Switchgears buses will be initiated between the ST unit bus and other ST unit bus by the protective relaying in the event of a fault within the relevant unit generator/ generator transformer / unit auxiliary transformer protection zone or if the relevant turbine is tripped.**
- (10) The MV auxiliary power system shall be designed to maintain the switchgear bus operating voltages (6.3 kV) within + 10% and - 10 % with $\pm 5\%$ generator/ system voltage variation. The medium voltage motors shall be rated 6 kV; the auxiliary system voltage variation shall correspond to + 10% and 5% of the motor rated voltage. The MV system will be designed to maintain more than 80% voltage at the motor terminals during starting of the largest motor and at least 90% switchgear bus voltage during normal operation.
- (11) An Auxiliary Power Control, that is operation and monitoring of auxiliary equipment and system, shall be performed from the Central Control Room through VDU based control.

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8.3.3 Low Voltage Auxiliary Power Distribution System

Reference Drawings

Key Single Line Diagrams

Reference Specifications

- E - 001 : General Electrical Specification
- E - 203 : Power Distribution Center Transformers
- E - 204 : Power Distribution Center Switchgears
- E - 205 : Motor Control Centers
- E - 223 : Emergency Diesel Engine Generator

Function and Design Requirements

- (1) The function of this system is to provide low voltage (400 V, 3 phase, 3 wire and 400/230 V, 3 phase, 4 wire) AC auxiliary power to loads consisting of low voltage motors, lighting, general low voltage distribution panels, etc.
- (2) The low-voltage (LV) auxiliary power system will furnish the normal power requirements of all the plant AC auxiliary loads up to 150 kW. The LV system will supply normal power to 3-phase as well as single phase AC loads; the vital electrical loads in the plant will be supplied from the 400 V essential service buses and /or from 230 V AC uninterruptible power supplies and /or from 220 V DC supplies.

For normal operation on the Steam Turbine Unit PDCs, the 400 V power will be obtained by PDC transformer. During startup and shutdown, the 400 V auxiliary power will be obtained by PDC transformers with stepping down the 500 kV off site power through the generator transformers and unit auxiliary transformers (UATs) with the generator circuit breakers open.

An "automatic transfer" of the 400 V unit essential buses will be initiated by the protective relaying in the event of a fault within the PDC transformer protection zone or if the 400 V PDC unit failed. This automatic transfer will be accomplished by first tripping the normal power supply breaker, then starting the EDG and closing the emergency

supply breaker connecting to the other unit auxiliary bus. A "fast" transfer scheme shall be used where necessary.

- (3) The LV auxiliary power system shall be designed to maintain the switchgear operating voltages (400 V) within + 10% and - 15% with $\pm 5\%$ generator/system voltage variation. The low voltage motors shall be rated 380 V, the auxiliary system voltage variation shall correspond to + 10% and - 5% of the motor rated voltage. The LV system shall be designed to maintain more than 80% voltage at the motor terminals during starting of the largest motor and at least 90% voltage at the motor terminals during normal operation.
- (4) Auxiliary power control shall also capable to be performed from the Central Control Room through VDU based control.

8.3.4 Station DC System

Reference Drawings

Key Single Line Diagram

Reference Specifications

E - 220 : Station Batteries and Chargers.

E - 221 : DC Distribution Panels

Function and Design Requirements

- (1) The station DC system shall, provide highly reliable DC power for the uninterruptible power supplies, emergency motors, emergency lighting, circuit breaker control and plant instrumentation and control.
- (2) The system shall consist of 220 V battery and 24 V battery (if required), associated battery chargers, DC switchgear, distribution panels, instrumentation and cabling for DC power and control circuits. This system shall be designed on the unit basis.
- (3) The DC system shall be ungrounded to enhance its reliability. It will be provided with ground detection for early detection of single ground. The batteries shall be sized to permit orderly emergency shutdown of the unit as well as for the plant control, instrumentation and emergency lighting loads, etc.

8.3.5 Uninterruptible Power Supply (UPS)

Reference Drawing

Key Single Line Diagrams

Reference Specification

E - 222 : Uninterruptible Power Supply

Function and Design Requirements

- (1) The 230 V, single phase, 50 Hz uninterruptible power supply system (UPS) shall furnish power for those electrical loads which require highly reliable, regulated and "clean" (with a minimum of spikes, electrical noise and harmonic distortion) low voltage AC power. The UPS shall be available continuously regardless of system disturbances.
- (2) The UPS shall operate in the "Reverse Mode" i.e. normally it shall convert the 220 V DC from the UPS rectifier/plant battery to 230 V AC, but in case of loss of input DC or loss of inverter output it shall automatically transfer to the alternate, regulated AC. The rectified DC in the UPS shall be backed by the plant DC battery.
- (3) The UPS shall have an auctioneering diode circuit at its DC input to isolate its DC from the station battery DC system. This shall also prevent connecting the station DC loads to the UPS rectifier.
- (4) The UPS rectifier shall be isolated from the AC power system by a shielded isolating transformer.
- (5) The static transfer switch will automatically transfer back and forth to the normal AC depending upon the status and condition of the normal (inverter) AC.

8.3.6 Emergency Power Supply System

Reference Drawing

Key Single Line Diagrams

Reference Specifications

E - 223 : Emergency Diesel Engine Generator

Function and Design Requirements

- (1) One (1) common emergency diesel engine generator supply system for Block 2, block 3 & 4 shall be furnished. The system shall include a diesel engine generator, control and distribution switchgear and power supply to the Essential PDCs. ADD1
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- (2) The diesel engine generator can be started automatically upon failure of the normal power supply to the Essential Service PDCs. This shall ensure that power is available for the safe shutdown of the unit when the unit is tripped and the normal AC power supply is lost.
- (3) The diesel generating set shall be designed to:
 - a. Start automatically or manually without any external electrical power.
 - b. Deliver full output within 10 seconds after starting.
- (4) Synchronizing facilities and associated circuit breakers shall provide a suitable unit load for periodic exercising of the diesel generator.
- (5) Provisions for remote starting of the diesels and operation of the output breakers shall be provided on the Central Control Room.

8.3.7 Grounding and Lightning Protection System

Reference Specifications

E - 240 : Outdoor and Substructure Grounding

E - 241 : Building and Equipment Grounding and Lightning Protection

Function and Design Requirement

- (1) The grounding system shall provide, a low impedance path between fault points and the neutral of the supply transformers so that adequate current can flow to ensure rapid action by the protective relays. The grounding system shall have adequate current carrying capacity to

prevent damage from overheating or sparking and limit the ground potential during faults to acceptable values.

- (2) The grounding system shall prevent injury to personnel during faults by limiting potential gradients to safe values.
- (3) The grounding system shall dissipate lightning and switching surge currents.
- (4) Since the grounding system is electrically connected to the station piping through the earth, the effect of the grounding system on the corrosion of station piping shall be considered.
- (5) The grounding system shall be divided into three (3) interconnected parts
 - a. The outdoor grounding system to control potential gradients during ground faults and dissipates the earth current during transmission faults. It consists of a system of buried cables installed near equipment and interconnecting metal structures. Lightning rods are installed in the grounding system where needed to help dissipate lightning and switching surge currents.
 - b. The building and equipment grounding system provides a low impedance path for ground fault currents. In addition, the indoor grounding system includes connections to the building steel to limit the potential differences between equipment to safe values.
 - c. The lightning protection system consists of a system of air terminals and down conductors to intercept lightning discharges large enough to cause structural damage to power plant buildings. Substation structures shall be protected by a system of aerial rods. The lightning current shall be then dissipated by the outdoor grounding system.
- (6) The grounding system shall be designed in accordance with IEEE 665 for Generating Station and IEEE 80 for Substation.
- (7) The outdoor and substructure grounding system shall be designed to interconnect all metallic structures in the power plant area and substation. All marine works shall be connected to the grounding system.

- (8) The safety ground shall control potential differences between equipment and building steel during ground faults. The safety ground consists of visible connections from the equipment frame to nearby building steel. Cable raceways shall be connected directly to building steel or to another section of raceway which is connected to building steel by jumpers. The building steel shall be connected to the outdoor and substructure grounding system. In addition to limiting potential differences, the safety ground provides a back-up earth return path.

8.3.8 Lighting System

Reference Specification

E - 270 : Lighting

Function and Design Requirements

- (1) The operation and maintenance of a power station and substation requires an adequate level of glare free light throughout. Illumination levels must be chosen to provide enough light for the safety of the workers and to permit easy maintenance of equipment. The lighting system must be economical to maintain and operate.
- (2) The lighting system shall be designed to use the most efficient, lowest maintenance cost light sources which provide satisfactory color rendition. High-pressure sodium lamps shall be used for high bay areas and fluorescent lamps shall be used for low bay areas. Fluorescent lamps shall be used for office areas and the control room.
- (3) The Stack shall be provided with approved aircraft obstruction lighting. Because of the importance of these lights, two (2) power feeds shall be provided, one (1) from an essential service bus. The aircraft obstruction lighting shall be designed to meet International Civil Aviation Organization (ICAO) standards and the specific local requirements, such as the Direktorat Jenderal Perhubungan Udara Regulation.
- (4) Most of the lighting shall be fed from 400/230 V lighting panels conveniently located throughout the plant. A small number of lamps shall be incandescent and fed from the 220 V DC battery to provide emergency lighting.

- (5) All equipment for the lighting system shall be manufactured by an approved Indonesian manufacturer, or if not so manufactured, it shall be readily available from suppliers in Indonesia.

8.3.9 Communications System

Reference Specification

E-271 : Plant Communication System.

Function and Design Requirements

- (1) The communications system :
- a. The private automatic branch exchange (PABX) shall be provided for both in plant and substation communications system and for gaining access to the public telephone system including the connection to the existing PABX. This system also shall be supplemented by self-contained amplified telephone system which shall be also used for startup, maintenance, and testing of plant equipment.
 - b. A public address system with handset stations located throughout the plant including the substation shall be provided. The public address system shall also have provisions to be used as an intercom.
- (2) All equipment and its components shall be of a standard design proven in similar applications. All equipment for the communications system shall be compatible with equipment in use in Indonesia and readily available from suppliers in Indonesia.

8.3.10 Wiring and Cabling Systems and Raceway System

Reference Specifications

E - 230 : Cables and Wires
E - 231 : Cable Tray System
E - 232 : Conduits and Fittings
E - 233 : Underground Electrical Ducts
S - 600 : Drainage and Duct Works

Function and Design Requirements

The function of these systems are to provide a safe, reliable and functional area-wide power plant and substation electrical systems. Several important features incorporated in the design of these systems to assure safety and reliability will be:

(1) Wiring and Cabling Systems

- a. Medium voltage power cables shall have shields over the conductor insulation with the shield effectively grounded.
- b. The cables shall be designed to withstand the maximum estimated short circuit current at the feed breaker for a period of time equal to the sum of the relaying and feed breaker operating times.
- c. Cables shall be color coded and numbered for consistency and clear identification.
- d. Cables belonging to different voltage groups shall be segregated to avoid mutual interference.
- e. Instrument cables shall be shielded, and the shields shall be effectively grounded.
- f. The total voltage drop in feeders or branch circuits shall not exceed 3% of bus nominal voltage.

(2) Raceway System

The raceway system shall include conduits, cable tray system underground electrical ducts and trenches.

a. Conduits

All conduits, conduit boxes and fittings shall be hot-dip galvanized. Conduits and cable tray system shall be electrically continuous. For outdoor installation, all boxes shall be galvanized and weatherproof, and shall have fixing lugs and the cover shall be provided with gasket of approved materials. Flexible metallic conduit with PVC sheath shall be used where relative movement is required between the conduit and connection provided on the outside to maintain ground continuity.

b. Cable Tray System

Trays for power, control and instrumentation cables shall be of galvanized sheet steel construction. Cable trays shall be separated and segregated by the voltage level and by their functions (power, control, communication and instrumentation) to avoid interference between the cables and circuits they will be carrying. If different type of power cables have to be run together; the control, communications and instrumentation trays shall be run below the power cable trays to avoid risk of fire damage in case of a fire in power cable tray at the floor and wall penetrations. Cable trays shall be segregated on a unit basis where possible.

c. Ducts and Trenches

Power and control cable runs between the plant and outlying areas shall be installed in underground concrete-lined trenches in which cable is supported by cable tray attached to the sides.

8.3.11 Clock System

Reference Specification

I- 200 : Plant Monitoring System

Function and Design Requirements

- (1) A master and slave clock system with the satellite communication system shall be provided. The master clock shall be located in the electronic equipment room and operated with the 220 V DC system. All monitoring system internal clocks shall be connected and synchronized to this clock.
- (2) Slave clocks shall be provided in the certain power plant areas, such as: Steam Turbine Building, Central Control Building, offices and conference rooms.

8.3.12 Cathodic Protection System

Reference Specification

E - 260 : Circulating Water System Cathodic Protection

E - 261 : Cathodic Protection for Field Piping and Buried Structures

Function and Design Requirements

- (1) Cathodic Protection System shall be Impressed Current and Sacrificial Anode System, and shall take into account material composition of the protected plants, interference from the grounding system and other structures.
- (2) The Cathodic Protection shall be provided for circuiting water system buried piping and all embedded steel structures.
- (3) The Cathodic Protection shall be provided for intake sheet pile, travelling screen, submarine pipe and all steel structures immersed in seawater.
- (4) Impressed Current Type Cathodic Protection System shall be automatically controlled with provision for manual operation.
- (5) Impressed Current Type Cathodic Protection System shall include provision for alarm and interlocking in case of failures.

8.3.13 Substation

Reference Drawing

500 kV Substation Extension Single Line Diagram 1 (Block 2)

500 kV Substation Extension Single Line Diagram 2 (Block 3 & 4)

Reference Specifications

E-001	:	General Electrical Specification.
E-400	:	Substation and Appurtenances.
E-401	:	Air Insulated Bus Work.
E-440	:	Substation Protection Relaying.
E-460	:	Substation Control System.

Function and Design Requirements

- (1) The substation will connect the electrical output of the power plant to the PLN 500 kV Network.
- (2) The substation shall be conventional type and provide the connections of:

500 kV connection to the generator transformer of steam turbine units.

- (3) The connection from steam turbine generator transformer to switchyard shall be by 500 kV CGIB as shown on attached Plot Plan.
- (4) There shall also be control/power cable ducts provided for the control and power cables connecting among the power plant, substation and control building.

8.3.14 Instrumentation and Control

A Microprocessor based Distributed Control System (DCS) shall be an integral part of the control system to perform the centralized modulating and discrete control function, display and plant monitoring functions. Functionally, this system shall provide for the start-up, normal operation, emergency operation and shutdown, automatic and manual operation including alarming.

The overall I&C system will be designed to ensure the following as minimum:

- (a) Maximize safety for the equipment and plant personnel
- (b) Safe, reliable and efficient operation of plant under operating conditions
- (c) Maximize availability and efficiency of the plant
- (d) Maximize the compatibility with the existing control system
- (e) Consistent instrumentation and control philosophy including high degree of simplicity, operability and maintainability
- (f) Ability to operate under all specified conditions (Start-up, normal operation, disturbances and shutdown) automatically and/or by remote manual control from relevant control room
- (g) Malfunction of any component or loss of supply will lead to fail-safe condition
- (h) Provide accurate and reliable information to the plant operators and management
- (i) Apply most modern and proven control system
- (j) Uniform of I&C equipment to supplied for maintainability

Adequate redundancies and back-up control will be provided where necessary so that failure of single component will not contribute abnormal condition, degrade the performance or shorten the life of the main plant equipment and electrical system. Redundancies will be ensured in terms of providing redundant control equipment and communication equipment wherever

necessary. Two (2) out of three (3) sensor (transmitter, switches etc) philosophy will be also adopted for critical control loops and interlocking system.

Reference Specifications

I-001	:	Instrumentation and Controls
I-002	:	Local Control Panels
I-003	:	Local Mounted Instrument Panels
I-005	:	Distributed Control System
I-100	:	Plant Control System
I-200	:	Plant Management System

Function and Design Requirements

Plant Control including HRSG control, ST control, and BOP control shall be controlled and monitored through VDUs based on operator station which shall be located in the Central Control Room.

The substation control, including monitoring, and operation shall be done by the LDC ordering the substation control and/or central control room for execution.

Start and stop of a certain equipment of Common Auxiliary Plant such as Desalination Plant, Hydrogen Plant, Water Treatment Plant, Waste Water Treatment Plant, Chlorination Plant, Fire Fighting System, Ventilation and Air Conditioning System, Compressed Air System and Condenser Cleaning System etc. shall be operated from VDU on operator station which shall be located in Central Control Room. However, process of Common Auxiliary Plant shall be controlled and monitored from their own control panels located locally in local control rooms.

(Last page of G-001)